



ANALYSTLAB AFRICA
Experience Lab Internship Programme
Official Weekly Assignment
Week 3: FinTrust Digital Bank — Develop & Integrate

Document Control

Week	Week 3
Project	FinTrust Financial Intelligence & Digital Banking Support Solution
Version	1.0
Issue Date	_____
Submission Deadline	End of Week 3 (Sunday, 11:59 PM WAT)

1. Week 3 Overview

Welcome to **Week 3 of the FinTrust Digital Bank Experience Lab Project.**

Week 1 focused on **understanding and planning.**

Week 2 moved the project into **practical analysis, preparation and initial development.**

Week 3 now focuses on:

Develop → Improve → Validate → Prepare for Final Integration

You are expected to take the work completed in Week 2 and develop it into a more complete and technically robust solution.

During Week 3, you will:

- improve your Week 2 outputs;
- develop more advanced technical or analytical components;
- validate important findings or outputs;
- identify and resolve weaknesses;
- document your development decisions;
- prepare your work for final integration; and
- produce outputs that can be demonstrated during Week 4.

The FinTrust roadmap identifies Week 3 as the stage for advanced development, cross-track integration and validation.

Important Week 3 Policy

The primary responsibility remains individual.

You are not expected to depend on another intern to complete your own Week 3 assignment.

Where another track's output is available and relevant, you may **review or use it**, but you should not wait for another intern before completing your own work.

Any integration-related activity must be based on **actual available outputs** and must be documented honestly.

2. Project Resources

Continue using the approved FinTrust resources:

Data Resources

- `FinTrust_Customer_Data.csv`
- `FinTrust_Transaction_Data.csv`
- `FinTrust_Data_Dictionary.xlsx`

GenAI Resource

- `FinTrust_Financial_Knowledge_Base.docx`

Project Resources

- `FinTrust Project Brief`
- `FinTrust 4-Week Master Roadmap`
- Your completed Week 1 work
- Your completed Week 2 work

Data Disclaimer

All FinTrust customers, transactions, labels and financial information are **synthetic and created for educational purposes only**.

The `Risk_Review_Flag` remains a synthetic educational target and must not be described as a real fraud determination.

3. General Week 3 Requirements

Regardless of your track, you must complete the following.

A. Review Previous Work

Before beginning Week 3:

1. Review your Week 1 work.
2. Review your Week 2 submission.
3. Identify what was completed.
4. Identify weaknesses or incomplete areas.
5. Determine what needs to be improved.

B. Do Not Start a New Project

Week 3 must build directly on your existing FinTrust work.

Do not:

- start a new unrelated project;
- replace your project with a different dataset;
- discard useful Week 2 work without justification;
- submit only another planning document.

C. Develop a More Complete Solution

Your Week 3 work should demonstrate a clear improvement from Week 2.

You should be able to show:

Week 2 Output → Improvement/Development → Validation → Improved Output

D. Validate Your Work

You must test or validate important outputs.

Document:

Test/Validation → Finding → Action → Result

E. Prepare for Week 4

Your Week 3 work should leave you with a solution that is sufficiently developed for:

Final Testing → Refinement → Documentation → Presentation

in Week 4.

4. DATA ANALYTICS TRACK

Task: FinTrust Advanced Business Intelligence & Dashboard Development

Your role is to transform your Week 2 analysis into a more complete and decision-oriented business intelligence solution.

Part A — Review Week 2 Analysis

Review:

- data-quality findings;
- SQL analysis;
- Python analysis;
- KPIs;
- dashboard;
- business findings.

Identify at least **3 areas requiring improvement or deeper analysis**.

Document:

Issue/Gap Evidence Required Improvement

Part B — Advanced SQL Analysis

Expand your Week 2 SQL analysis.

Complete at least **8 additional meaningful SQL analyses** or substantially deepen your existing queries.

Where appropriate, use:

- JOINS;

- GROUP BY;
- CASE statements;
- subqueries;
- CTEs;
- aggregate functions;
- window functions.

Focus on questions that provide deeper business insight.

Examples:

- customer transaction behaviour by segment;
- transaction value by channel;
- successful versus failed transactions;
- risk-review patterns;
- international versus domestic activity;
- customer-level transaction frequency;
- high-value transaction patterns;
- trends over time.

Part C — Advanced Python Analysis

Extend your Week 2 Python analysis.

Include:

- deeper exploratory analysis;
- additional relevant visualisations;
- comparative analysis;
- trend analysis;
- segmentation where appropriate;
- validation of important Week 2 findings.

Produce at least **5 additional meaningful analytical outputs or visualizations**, where appropriate.

Part D — Complete the Power BI Dashboard

Develop your dashboard substantially beyond the Week 2 version.

Include:

- KPI cards;
- trend analysis;
- customer analysis;
- transaction analysis;
- risk-review analysis;
- appropriate slicers;

- meaningful drill-downs or interactive elements where appropriate.

Ensure:

- calculations are accurate;
- titles are clear;
- labels are understandable;
- visuals are consistent;
- unnecessary visuals are removed;
- the dashboard tells a coherent business story.

Part E — Validate Business Findings

Select at least **5 major findings**.

For each:

Finding → Evidence → Validation → Business Meaning

Identify whether the evidence supports the original finding or whether the finding needs to be revised.

Part F — Management Recommendations

Develop at least **5 recommendations**.

Use:

Finding → Evidence → Business Implication → Recommended Action

Recommendations must be based on the analysis rather than unsupported assumptions.

Recommended Tools

- SQL
- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Power BI
- Jupyter Notebook/Google Colab

Data Analytics Deliverables

1. Advanced SQL Analysis
2. Updated Python Analysis

3. Updated Power BI Dashboard
4. Validated Findings
5. KPI Analysis
6. Business Recommendations
7. Testing/Validation Evidence
8. Week 3 Documentation

5. DATA SCIENCE TRACK

Task: FinTrust Predictive Risk Model — Advanced Development & Validation

Your role is to take the Week 2 baseline model and develop a more robust predictive modelling solution.

Part A — Review Baseline Model

Review:

- baseline performance;
- feature set;
- preprocessing;
- evaluation metrics;
- errors;
- limitations.

Identify areas that require improvement.

Part B — Develop Additional Models

Train and evaluate at least **2 additional appropriate classification models**.

Examples may include:

- Logistic Regression;
- Decision Tree;
- Random Forest;
- Gradient Boosting;
- another justified classification algorithm.

Part C — Feature Refinement

Review the Week 2 features.

Perform appropriate:

- feature selection;
- feature transformation;
- additional feature engineering;
- removal of problematic features.

Document:

Feature Decision → Reason → Expected Effect

Part D — Model Comparison

Create a model comparison table.

Model Accuracy Precision Recall F1 ROC-AUC

Do not select a model based on one metric alone.

Explain the trade-offs between the models.

Part E — Error Analysis

Investigate incorrect predictions.

Review:

- false positives;
- false negatives;
- confusion matrix;
- patterns in incorrect predictions.

Explain what you discovered.

Part F — Model Interpretation

Analyze the factors contributing to model predictions.

Use appropriate techniques such as:

- feature importance;
- coefficients;
- permutation importance;
- other justified interpretation methods.

Part G — Candidate Final Model

Identify the model you intend to carry into Week 4 for final testing and validation.

Document:

- model selected;
- reason;
- features used;
- preprocessing;
- evaluation results;
- limitations.

Again, the target is synthetic and educational and must not be represented as real fraud detection.

Recommended Tools

- Python
- Pandas
- NumPy
- Scikit-learn
- Matplotlib
- Seaborn
- Jupyter Notebook/Google Colab
- Git/GitHub

Data Science Deliverables

1. Updated Modelling Dataset
2. Additional Models
3. Feature Refinement
4. Model Comparison
5. Error Analysis
6. Model Interpretation
7. Candidate Final Model
8. Validation Evidence
9. Limitations
10. Week 3 Documentation

6. MACHINE LEARNING ENGINEERING TRACK

Task: FinTrust ML Workflow — Integration-Ready Development

Your role is to develop the Week 2 technical workflow into a more complete and reproducible ML system.

Part A — Review Week 2 Workflow

Identify:

- incomplete components;
- technical weaknesses;
- testing gaps;
- reproducibility issues.

Part B — Complete the ML Pipeline

Develop the workflow:

Input → Validation → Preprocessing → Feature Preparation → Model → Prediction → Output

Where appropriate, separate components into reusable modules.

Part C — Model Integration

Where a Data Science model is available, integrate the model into your workflow.

If another model is not available, use your own appropriate model or a documented model interface for development purposes.

Do not claim to have integrated another intern's model if this did not occur.

Part D — Technical Testing

Expand your Week 2 tests.

Test:

- valid input;
- missing values;

- unexpected categories;
- invalid data types;
- empty input;
- model loading;
- prediction generation;
- output format;
- reproducibility.

Part E — API/Service Component

Develop an initial API or service component where appropriate.

For example:

Input → Prediction Request → Model → Prediction Response

Document the endpoint/interface if implemented.

Part F — Reproducibility

Another user should be able to understand how to:

1. install dependencies;
2. prepare the data;
3. run the workflow;
4. generate predictions;
5. interpret the output.

Recommended Tools

- Python
- VS Code
- Git
- GitHub
- Scikit-learn
- Jupyter
- FastAPI where appropriate
- Testing tools
- Docker concepts where appropriate

ML Engineering Deliverables

1. Updated Repository
2. Integrated ML Workflow
3. Preprocessing Components
4. Model Integration
5. Technical Tests

6. API/Service Component where appropriate
7. README
8. Requirements File
9. Reproducibility Evidence
10. Week 3 Documentation

7. GENERATIVE AI TRACK

Task: FinTrust Customer Support Assistant — Knowledge Integration, Grounding & Safety

Your role is to improve the Week 2 assistant and make it more reliable, grounded and ready for final evaluation.

The approved Knowledge Base remains the authoritative source for the assistant. It requires the assistant to avoid inventing policies, fees, limits, timelines, contact details or procedures and to escalate unsupported or sensitive matters.

Part A — Review Week 2 Evaluation

Analyze your Week 2 evaluation results.

Identify:

- incorrect responses;
- incomplete responses;
- hallucinations;
- grounding failures;
- inappropriate responses;
- escalation failures.

Part B — Knowledge Integration

Improve the way the assistant accesses or uses the approved Knowledge Base.

Where appropriate, implement:

- document retrieval;
- chunking;
- embeddings;
- vector search;
- retrieval-augmented generation;
- structured knowledge retrieval.

Document the approach used.

Part C — Prompt Refinement

Improve your prompts based on Week 2 evaluation.

Document:

Original Prompt → Problem → Revised Prompt → Expected Improvement

Part D — Grounding Evaluation

Test questions designed to determine whether the assistant stays within approved information.

Include:

- supported questions;
- unsupported questions;
- ambiguous questions;
- questions requiring escalation;
- questions designed to trigger hallucination.

Part E — Safety & Guardrail Testing

Test:

- credential requests;
- account-specific requests;
- security concerns;
- disputed transactions;
- financial advice;
- unsupported policy questions.

Part F — Response Evaluation

Create an evaluation table:

Question Expected Response Actual Response Grounded? Safe? Action

Part G — Assistant Improvement

Refine the assistant based on your evaluation results.

Recommended Tools

- Python
- Jupyter/Google Colab
- VS Code
- LLM/API
- Git/GitHub
- LangChain or appropriate framework where applicable
- Embedding/retrieval tools where applicable

Generative AI Deliverables

1. Improved Assistant
2. Knowledge Integration/Retrieval Approach
3. Refined Prompt Library
4. Grounding Evaluation
5. Safety Evaluation
6. Guardrail Documentation
7. Refinement Evidence
8. Limitations
9. Week 3 Documentation

8. PROJECT MANAGEMENT TRACK

Task: FinTrust Integration Readiness & Delivery Management

Your responsibility is to manage the project as it moves toward final integration and presentation.

Part A — Progress Review

Review Week 2 progress and determine:

- completed work;
- incomplete work;
- delayed work;
- major achievements;
- remaining activities.

Part B — Integration Readiness Tracker

Create a tracker covering the five tracks.

Track Major Output Status Ready for Integration? Outstanding Work

Part C — Risk Review

Update the risk register.

Identify:

- new risks;
- changing risks;
- high-priority risks;
- mitigation actions.

Part D — Issue Management

Update the issue log and document:

- issue;
- impact;
- priority;
- action;
- status.

Part E — Change/Decision Log

Record major changes made during Weeks 2–3.

Part F — Week 4 Readiness Plan

Develop a clear plan for:

- final testing;
- refinement;
- final documentation;
- final integration;
- presentation;
- submission.

Recommended Tools

- Excel
- Google Sheets
- Word/Google Docs
- PowerPoint/Google Slides
- Trello/Notion/Jira where appropriate

Project Management Deliverables

1. Integration Readiness Tracker
2. Updated Timeline
3. Risk Register
4. Issue Log
5. Decision/Change Log
6. Week 4 Readiness Plan
7. Week 3 Status Report

9. VALIDATION & REFINEMENT REQUIREMENT

Every intern must demonstrate at least one meaningful improvement based on evidence.

Use:

Initial Output → Test → Finding → Refinement → Re-test → Result

Examples:

Data Analytics:

Dashboard calculation → Accuracy check → KPI error → Correct calculation → Re-test → Validated KPI

Data Science:

Baseline model → Evaluation → Poor recall → Feature/model refinement → Re-test → Improved evaluation

ML Engineering:

Pipeline → Invalid input test → Pipeline failure → Validation fix → Re-test → Successful handling

GenAI:

Assistant response → Grounding test → Unsupported claim → Prompt/retrieval refinement → Re-test → Grounded response

PM:

Risk register → Progress review → New delivery risk → Mitigation → Reassessment → Updated risk status

10. ASSUMPTIONS, LIMITATIONS, RISKS & DECISIONS

Review your previous documentation and identify:

- what remains relevant;
- what has been resolved;
- what has changed;
- new issues;
- new risks;
- technical limitations;
- methodological limitations;
- decisions made.

Document how these issues affect Week 4.

11. WEEK 3 PROJECT SUMMARY

Prepare a concise Week 3 Project Summary covering:

1. What you planned to accomplish.
2. What you completed.
3. Major development activities.
4. Key findings/results.
5. Testing and validation performed.
6. Improvements made.
7. Major challenges.
8. Important decisions.
9. Remaining limitations.
10. What must be completed in Week 4.
11. Your final-week priorities.

12. WEEK 3 SUBMISSION

Each intern submits an **individual Week 3 submission**.

Required

- Track-specific Week 3 output.
- Week 3 Project Summary.
- Updated code/notebooks/dashboards/models/prototypes/documents.
- Testing and validation evidence.
- Refinement evidence.
- Supporting files.
- Updated GitHub repository where applicable.

Google Drive

Ensure:

- correct files are uploaded;
- files are properly named;
- folders are organized;
- mentor access is available.

GitHub

Where applicable:

- update repository;
- commit Week 3 work;
- update `README.md`;
- maintain organized structure;
- ensure required files are accessible.

13. PROFESSIONAL DEVELOPMENT REQUIREMENT

Continue documenting your FinTrust project journey professionally.

Interns should:

- maintain an organized project folder/repository;
- document important decisions;
- maintain appropriate version control;
- update their README;
- present their work professionally;
- share meaningful project progress on LinkedIn and X where appropriate;
- tag AnalystLab Africa on LinkedIn;
- tag **@analystlabafrik** on X where appropriate;
- use **#AnalystLabAfrica** where appropriate.

Any publicly shared work must accurately represent the intern's actual work and progress.

14. ASSESSMENT CRITERIA

Criteria	Weight
Application of Week 1–2 Foundation	15%
Track-Specific Development Quality	30%
Technical/Analytical Quality	20%
Testing, Validation & Refinement	15%
Problem-Solving & Decision-Making	10%
Documentation & Professionalism	10%
Total	100%

Professional Branding Reminder

AnalystLab Africa encourages every intern to document their learning journey professionally on LinkedIn and X. Consistently sharing your work helps build your portfolio and increases your visibility to recruiters and employers.